MA-UY 2224 Data Analysis

Spring 2027 Course Syllabus

Instructor: Prof. Lucia Ferrante (lf1180@nyu.edu). Office hours Tue 2–4 PM, RH 621.

Lecture: Tuesday and Thursday, 12:30–1:45 PM, JAB 473. Lab: Thursday 3:00–3:50 PM, RH 300 (computer lab).

First day of classes: Monday, January 25, 2027. Last day of classes: Monday, May 10, 2027. Spring break: March 15–19, 2027.

Course description

An introduction to statistical reasoning with real data. Topics include exploratory data analysis, probability, sampling distributions, confidence intervals, hypothesis tests, and regression. Computation is done in R; no prior programming experience is assumed. Each week pairs a lecture with a hands-on lab in which you analyze a dataset and write up findings.

Learning outcomes

By the end of the course you will be able to summarize a dataset numerically and graphically, state and check the assumptions of a statistical procedure, compute and interpret confidence intervals and p-values, fit and diagnose a simple linear regression, and communicate a quantitative analysis in writing to a non-specialist reader.

Grading

Weekly Labs	30%	ten labs on Gradescope; lowest lab score dropped
Project	20%	proposal (5%) and final report (15%)
Midterm	20%	
Final Exam	30%	cumulative

Labs are due on Thursdays at 11:59 PM on Gradescope, one week after they are assigned. Grades are curved only at the end of the term, and only upward.

Lab schedule

Lab 1	Data and R basics	due Thu Feb 4
Lab 2	Graphics with ggplot	due Thu Feb 11
Lab 3	Numerical summaries	due Thu Feb 18
Lab 4	Probability simulation	due Thu Feb 25
Lab 5	Sampling distributions	due Thu Mar 4
Lab 6	Confidence intervals	due Thu Mar 11
Lab 7	Hypothesis tests	due Thu Mar 25
Lab 8	Two-sample comparisons	due Thu Apr 1
Lab 9	Simple regression	due Thu Apr 8
Lab 10	Regression diagnostics	due Thu Apr 15

Exams and project milestones

Midterm: Tuesday, March 9, in class. Covers material through Lab 5.

Project proposal due Friday, March 26, 11:59 PM on Gradescope (one page: dataset, question, planned method).

Project report due Friday, May 7, 11:59 PM on Gradescope (six to eight pages plus code appendix).

Final Exam: Wednesday, May 12, 10:00 AM–12:00 PM, room assigned by the registrar.

Weekly topics

Week 1 (Jan 26, 28): What is data? Variables, cases, the R environment.

Week 2 (Feb 2, 4): Visualizing distributions; histograms, boxplots, scatterplots.

Week 3 (Feb 9, 11): Numerical summaries; mean, median, spread, outliers.

Week 4 (Feb 16, 18): Probability rules; conditional probability; independence.

Week 5 (Feb 23, 25): Random variables; binomial and normal distributions.

Week 6 (Mar 2, 4): Sampling distributions; the central limit theorem.

Week 7 (Mar 9, 11): Midterm; confidence intervals for a mean.

Week 8 (Mar 23, 25): Hypothesis testing logic; p-values and errors.

Week 9 (Mar 30, Apr 1): Comparing two groups; paired designs.

Week 10 (Apr 6, 8): Correlation and simple linear regression.

Week 11 (Apr 13, 15): Regression inference and diagnostics.

Week 12 (Apr 20, 22): Categorical data; chi-square tests.

Week 13 (Apr 27, 29): Multiple regression preview; project work.

Week 14 (May 4, 6): Review; project presentations in lab.

Policies

Late labs: Labs may be submitted up to 24 hours late with a 20% penalty; nothing is accepted after 24 hours. The dropped lowest lab score is meant to absorb one missed lab.

Makeup exams: A makeup midterm is available only with documentation from the Student Health Center or the Moses Center, submitted within one week.

Attendance: Lab attendance is required; you may miss one lab without penalty.

Collaboration: You may discuss labs with classmates, but all code and writing must be your own. Project reports are individual.

Extra credit: None is offered in this course.